



Parul University

FACULTY OF ENGINEERING AND TECHNOLOGY
BACHELOR OF TECHNOLOGY

Mobile App Development Laboratory
(303105380)

6th SEMESTER (3RD YEAR)
COMPUTER SCIENCE & ENGINEERING DEPARTMENT

LAB MANUAL



CERTIFICATE

This is to certify that Mr./Ms **KUNAM MANI SAI CHARAN REDDY** with Enrollment no. **2303031250067** has successfully completed his/her laboratory experiments in **Mobile App Development Laboratory (303105380)** from the Department of Computer Science and Engineering (Big Data Analytics) during the academic year 2025-26.

Date of Submission:

Staff In charge: Head of Department:

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Practical-1

Aim: Create a "Hello World" application

Xml file code(activity_main.xml):

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World 2"/>

</LinearLayout>
```

MainActivity.java code:

```
package com.example.myapplication;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });
}
```

Output:

Practical-2

Aim: Create an app with explicit intent

activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Retro Phone Booth"
        android:textAlignment="center"
        android:textStyle="bold"
        android:background="#a25654"
        android:textColor="#fff"
        android:textSize="40dp"/>

    <ImageView
        android:layout_width="wrap_content"
        android:layout_height="300dp"
        android:src="@drawable/tell" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:paddingRight="10dp"
        android:paddingLeft="10dp">

        <EditText
            android:id="@+id/eText"
            android:layout_width="match_parent"
            android:paddingRight="10dp"
            android:paddingLeft="10dp"
            android:layout_height="50dp"
            android:layout_gravity="center"
            android:hint="Enter Your Number"/>

        <Button
            android:id="@+id/callBtn"
```

```
android:layout_width="120dp"
android:layout_height="50dp"
android:text="Call Now"
android:layout_marginTop="10dp"
android:layout_gravity="center"
android:backgroundTint="#a25654"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="bottom"
    android:orientation="vertical">
```

```
<com.google.android.material.button.MaterialButton
    android:id="@+id/shareBtn"
    android:layout_width="100dp"
    android:layout_height="60dp"
    app:icon="@drawable/share_icon"
    app:iconSize="28dp"
    app:iconGravity="textEnd"
    android:contentDescription="share"
    android:layout_gravity="end"
    android:layout_marginBottom="20dp"
    android:backgroundTint="#a25654"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

MainActivity.java code:

```
package com.example.implicit_intent;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```



```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        //----- creating the button -----
        EditText editText;
        Button callBtn, shareBtn;

        //----- finding that view by id -----
        editText= findViewById(R.id.eText);
        callBtn = findViewById(R.id.callBtn);
        shareBtn = findViewById(R.id.shareBtn);

        //----- creating intent -----
        //this intent is for dial number
        Intent dialIntent = new Intent(Intent.ACTION_DIAL);
        Intent shareIntent = new Intent(Intent.ACTION_SEND);

        //----- code for call button click -----
        callBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                //---- extracting text from input field -----
                String number = editText.getText().toString();
                dialIntent.setData(Uri.parse("tel: "+ number));
                startActivity(dialIntent);
            }
        });

        // ----- code for share button click -----
        shareBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                shareIntent.setType("text/plain");
                shareIntent.putExtra(Intent.EXTRA_TEXT, "My Portfolio,
https://saketfolio.netlify.app/");
                startActivity(shareIntent);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
        {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
```

```
v.setPadding(systemBars.left, systemBars.top,  
systemBars.right, systemBars.bottom);  
return insets;  
});  
}  
}
```

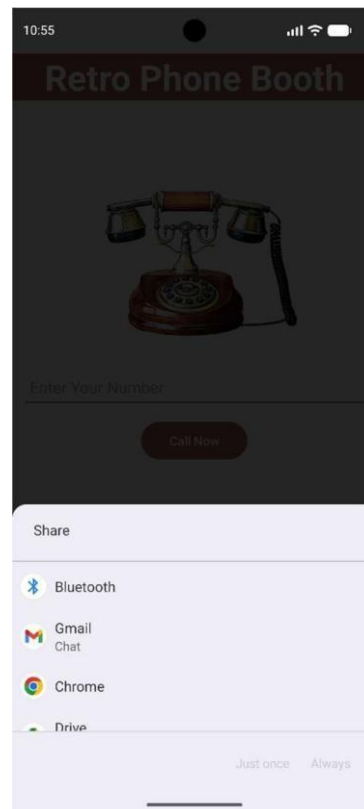
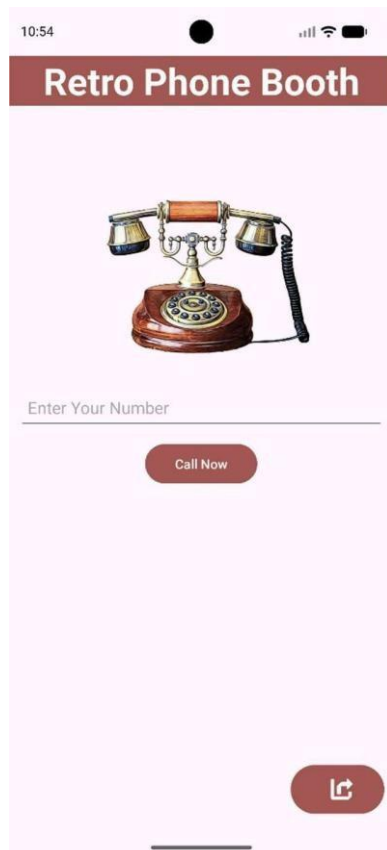
colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>  
<resources>  
    <color name="black">#FF000000</color>  
    <color name="white">#FFFFFFF</color>  
</resources>
```

strings.xml code:

```
<resources>  
    <string name="app_name">Implicit_Intent</string>  
</resources>
```

Output:



Practical-3

Aim: Create an app with two activities

activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:paddingLeft="12dp"
    android:paddingRight="12dp"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="First Screen"
        android:textSize="30dp"
        android:textAlignment="center"
        android:textStyle="bold"/>

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:src="@drawable/tom"/>

    <!-- <ImageView
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:src="@drawable/tom"/>-->

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="We Are Friends"
        android:textSize="20dp"
        android:textAlignment="center"
        android:layout_marginBottom="20dp"
        android:textStyle="bold"/>

    <EditText
        android:id="@+id/eText"
        android:layout_width="match_parent"
        android:layout_height="50dp"
```

```
android:textColor="@color/black"  
android:textSize="20dp"  
android:hint="Enter Your Name"/>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:gravity="bottom|center"  
    android:orientation="vertical">  
  
    <Button  
        android:id="@+id/nextBtn"  
        android:layout_width="120dp"  
        android:layout_height="60dp"  
        android:textStyle="bold"  
        android:textSize="18dp"  
        android:backgroundTint="#ff0000"  
        android:layout_marginBottom="100dp"  
        android:text="Next"/>  
</LinearLayout>
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.intent;  
  
import android.content.Intent;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.Button;  
import android.widget.EditText;  
  
import androidx.activity.EdgeToEdge;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;  
import androidx.core.view.WindowInsetsCompat;  
  
public class MainActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        EdgeToEdge.enable(this);  
        setContentView(R.layout.activity_main);  
  
        // calling button  
        Button nextBtn;  
        nextBtn = findViewById(R.id.nextBtn);
```

```
EditText eText;
eText = findViewById(R.id.eText);

// making intent
Intent inext;

// use inbuilt class to navigate btwn activity
inext = new Intent(MainActivity.this , SecondActivity.class);

//----- to share data from main act to second act -----
// adding click method to the button

nextBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

// passing the data as string and int
        String myname = eText.getText().toString();
        inext.putExtra("myName", myname);

        inext.putExtra("myName", "Shubhangi Jha");
// start activity of the intent named as inext
        startActivity(inext);

    }
});

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(
        systemBars.left + v.getPaddingLeft(),
        systemBars.top + v.getPaddingTop(),
        systemBars.right + v.getPaddingRight(),
        systemBars.bottom + v.getPaddingBottom()
    );
    return insets;
});
}
```

activity_second.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".SecondActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Second Screen"
        android:textSize="30dp"
        android:textStyle="bold"
        android:textAlignment="center"/>

    <TextView
        android:id="@+id/tName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Name: "
        android:textSize="25dp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"
        android:textAlignment="center"/>

    <!--<TextView
        android:id="@+id/tAge"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Age: "
        android:textSize="25dp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"
        android:textAlignment="center"/>
-->

</LinearLayout>
```

SecondActivity.java code

```
package com.example.intent;

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_second);

        // getting intent from main-activity
        Intent fromMain = getIntent();

        // storing the values in variable
        String name = fromMain.getStringExtra("myName").toString();
        // String sName = fromMain.getStringExtra("secondName").toString();
        // int age = fromMain.getIntExtra("age", 0);

        // calling text-view as id name
        TextView tName;
        TextView sName;
        //TextView tAge;

        // finding the view by id
        // tAge = findViewById(R.id.tAge);
        tName = findViewById(R.id.tName);

        // set this data in view
        tName.setText(name);
        //tAge.setText(age + "");

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
        {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

```
}  
}
```

activity_third.xml code:

```
<?xml version="1.0" encoding="utf-8"?>  
<androidx.constraintlayout.widget.ConstraintLayout  
    xmlns:android="http://schemas.android.com/apk/res/android"  
        xmlns:app="http://schemas.android.com/apk/res-auto"  
        xmlns:tools="http://schemas.android.com/tools"  
        android:id="@+id/main"  
        android:layout_width="match_parent"  
        android:layout_height="match_parent"  
        tools:context=".ThirdActivity">  
  
</androidx.constraintlayout.widget.ConstraintLayout>
```

ThirdActivity.java

```
package com.example.intent;  
  
import android.os.Bundle;  
  
import androidx.activity.EdgeToEdge;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;  
import androidx.core.view.WindowInsetsCompat;  
  
public class ThirdActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        EdgeToEdge.enable(this);  
        setContentView(R.layout.activity_third);  
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->  
        {  
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());  
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);  
            return insets;  
        });  
    }  
}
```

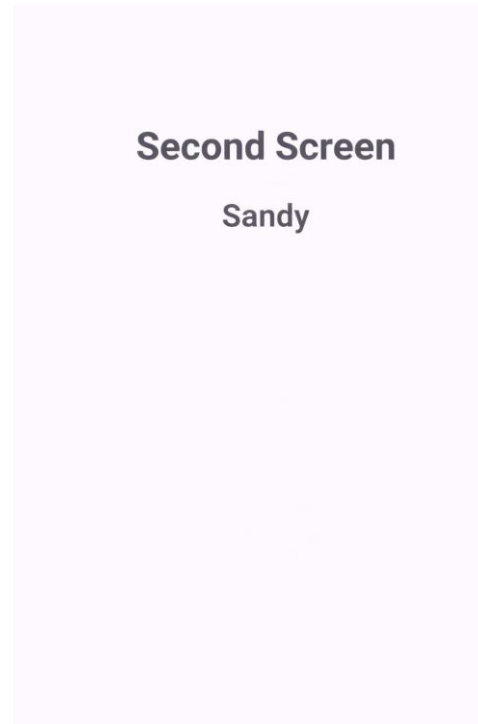
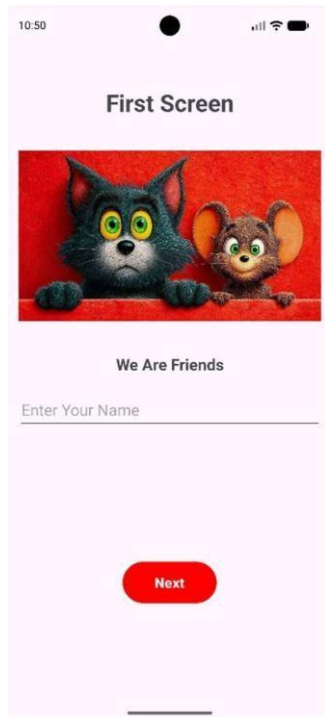
colors.xml code:


```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <color name="black">#FF000000</color>
  <color name="white">#FFFFFFF</color>
</resources>
```

strings.xml code:

```
<resources>
  <string name="app_name">Intent</string>
</resources>
```

OUTPUT



Practical-4

Aim: Create an app with explicit intent

activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Retro Phone Booth"
        android:textAlignment="center"
        android:textStyle="bold"
        android:background="#a25654"
        android:textColor="#fff"
        android:textSize="40dp"/>

    <ImageView
        android:layout_width="wrap_content"
        android:layout_height="300dp"
        android:src="@drawable/tell" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:paddingRight="10dp"
        android:paddingLeft="10dp">

        <EditText
            android:id="@+id/eText"
            android:layout_width="match_parent"
            android:paddingRight="10dp"
            android:paddingLeft="10dp"
            android:layout_height="50dp"
            android:layout_gravity="center"
            android:hint="Enter Your Number"/>

        <Button
```

```
android:id="@+id/callBtn"  
android:layout_width="120dp"  
android:layout_height="50dp"  
android:text="Call Now"  
android:layout_marginTop="10dp"  
android:layout_gravity="center"  
android:backgroundTint="#a25654"/>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:gravity="bottom"  
    android:orientation="vertical">
```

```
<com.google.android.material.button.MaterialButton  
    android:id="@+id/shareBtn"  
    android:layout_width="100dp"  
    android:layout_height="60dp"  
    app:icon="@drawable/share_icon"  
    app:iconSize="28dp"  
    app:iconGravity="textEnd"  
    android:contentDescription="share"  
    android:layout_gravity="end"  
    android:layout_marginBottom="20dp"  
    android:backgroundTint="#a25654"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

MainActivity.java code:

```
package com.example.implicit_intent;  
  
import android.content.Intent;  
import android.net.Uri;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.Button;  
import android.widget.EditText;  
  
import androidx.activity.EdgeToEdge;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;
```

```
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        //----- creating the button -----
        EditText editText;
        Button callBtn, shareBtn;

        //----- finding that view by id -----
        editText= findViewById(R.id.eText);
        callBtn = findViewById(R.id.callBtn);
        shareBtn = findViewById(R.id.shareBtn);

        //----- creating intent -----
        //this intent is for dial number
        Intent dialIntent = new Intent(Intent.ACTION_DIAL);
        Intent shareIntent = new Intent(Intent.ACTION_SEND);

        //----- code for call button click -----
        callBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                //---- extracting text from input field -----
                String number = editText.getText().toString();
                dialIntent.setData(Uri.parse("tel: "+ number));
                startActivity(dialIntent);
            }
        });

        // ----- code for share button click -----
        shareBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                shareIntent.setType("text/plain");
                shareIntent.putExtra(Intent.EXTRA_TEXT, "My Portfolio,
https://saketfolio.netlify.app/");
                startActivity(shareIntent);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
        {
```

```
Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
    });
}
```

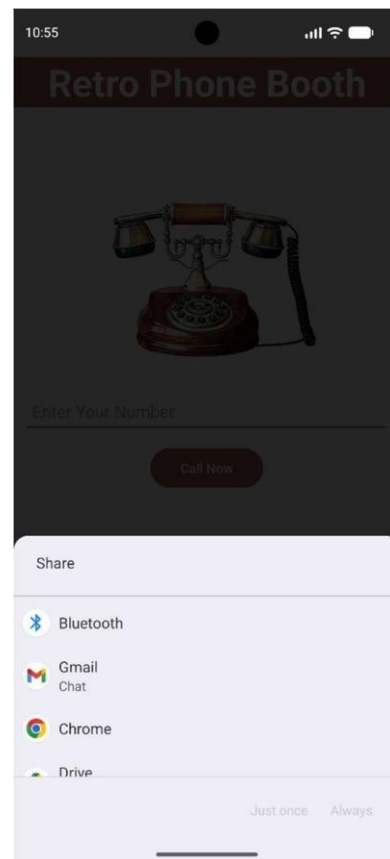
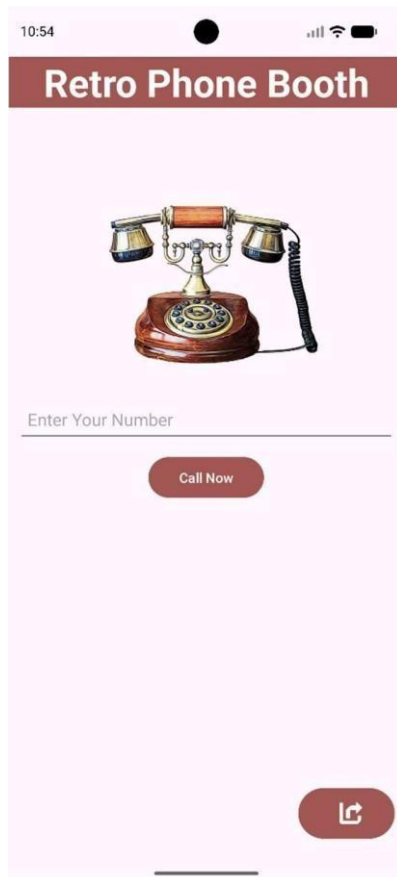
colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

strings.xml code:

```
<resources>
    <string name="app_name">Implicit_Intent</string>
</resources>
```

Output:



Practical-5

Aim: Build a basic calculator app.

Activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    android:orientation="vertical"
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="CALCULATOR"
        android:textSize="48sp"
        android:textAlignment="center"
        android:textStyle="bold"
        android:fontFamily="@font/sophiefancy"
        android:layout_marginTop="20dp"
        android:letterSpacing="0.07"
        />

    <EditText
        android:id="@+id/firstValue"
        style="@style/editText"
        android:hint="Enter 1st Value"
        android:layout_marginTop="100dp"
        />

    <EditText
        android:id="@+id/secValue"
        android:hint="Enter 2nd Value"
        style="@style/editText"
        android:layout_marginTop="60dp"
        />

    <TextView
```

```
android:id="@+id/ans"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:text="Result: "  
android:fontFamily="@font/sophiefancy"  
android:textAlignment="center"  
android:textSize="30sp"  
android:textStyle="bold"  
android:layout_marginTop="50dp"  
</>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:layout_marginTop="80dp"  
    android:padding="10dp"  
    android:orientation="horizontal">
```

```
<Button  
    android:id="@+id/addBtn"  
    style="@style/buttonView"  
    android:text="+" />
```

```
<Button  
    android:id="@+id/subBtn"  
    style="@style/buttonView"  
    android:text="-" />
```

```
</LinearLayout>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:layout_marginTop="10dp"  
    android:padding="10dp"  
    android:orientation="horizontal">
```

```
<Button  
    android:id="@+id/mulBtn"  
    style="@style/buttonView"  
    android:text="*" />
```

```
<Button  
    android:id="@+id/divBtn"  
    style="@style/buttonView"  
    android:text="/" />
```

```
</LinearLayout>
```

```
<TextView  
    android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
android:text="@string/developer "
android:textAlignment="center"
android:layout_marginTop="30dp"
android:textSize="20dp"
android:fontFamily="@font/sophiefancy"
android:layout_gravity="end"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.calculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // creating edit text
        TextView ans;
        EditText firstValue, secValue;

        // finding that edit text
        ans = findViewById(R.id.ans);
        firstValue = findViewById(R.id.firstValue);
        secValue = findViewById(R.id.secValue);

        // creating button
        Button addBtn, subBtn, mulBtn, divBtn;

        // finding these buttons
        addBtn = findViewById(R.id.addBtn);
        subBtn = findViewById(R.id.subBtn);
        mulBtn = findViewById(R.id.mulBtn);
```



```
divBtn = findViewById(R.id.divBtn);

// code for add button click
addBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        String value1 = firstValue.getText().toString();
        String value2 = secValue.getText().toString();

        if(value1.isEmpty() || value2.isEmpty()){
            Toast.makeText(MainActivity.this, "Enter Value",
Toast.LENGTH_SHORT).show();
            ans.setText("Result: ");
            return;
        }else {
            int num1 = Integer.parseInt(value1);
            int num2 = Integer.parseInt(value2);

            int result = num1 + num2;
            ans.setText("Result: "+ result);
        }
    }
});

// code for sub button click
subBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String value1 = firstValue.getText().toString();
        String value2 = secValue.getText().toString();

        if(value1.isEmpty() || value2.isEmpty()){
            Toast.makeText(MainActivity.this, "Enter Value",
Toast.LENGTH_SHORT).show();
            ans.setText("Result: ");
            return;
        }else {
            int num1 = Integer.parseInt(value1);
            int num2 = Integer.parseInt(value2);

            int result = num1 - num2;
            ans.setText("Result: "+ result);
        }
    }
});

// code for mul button click
mulBtn.setOnClickListener(new View.OnClickListener() {
```

```
@Override
public void onClick(View v) {
    String value1 = firstValue.getText().toString();
    String value2 = secValue.getText().toString();

    if(value1.isEmpty() || value2.isEmpty()){
        Toast.makeText(MainActivity.this, "Enter Value",
Toast.LENGTH_SHORT).show();
        ans.setText("Result: ");
        return;
    }else {
        int num1 = Integer.parseInt(value1);
        int num2 = Integer.parseInt(value2);

        int result = num1 * num2;
        ans.setText("Result: "+ result);
    }
}

});
```

```
// code for div button click
divBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String value1 = firstValue.getText().toString();
        String value2 = firstValue.getText().toString();

        if(value1.isEmpty() || value2.isEmpty()){
            Toast.makeText(MainActivity.this, "Enter Value",
Toast.LENGTH_SHORT).show();
            ans.setText("Result: ");
            return;
        }else {
            int num1 = Integer.parseInt(value1);
            int num2 = Integer.parseInt(value2);

            int result = num1 / num2;
            ans.setText("Result: "+ result);
        }
    }
});
```

```
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
```

```
Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
    });
    }
}
```

colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

Strings.xml code:

```
<resources>
    <string name="app_name">Calculator</string>
    <string name="developer">Developed by @ Shubhangi Jha</string>
</resources>
```

Styles.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    // style for edit text view

    <style name="editText">
        <item name="android:textSize">20dp</item>
        <item name="android:layout_width">300dp</item>
        <item name="android:padding">10dp</item>
        <item name="android:layout_gravity">center</item>
        <item name="android:layout_height">50dp</item>
        <item name="android:inputType">number</item>
    </style>

    // style for button view

    <style name="buttonView">
        <item name="android:layout_height">wrap_content</item>
        <item name="android:layout_width">wrap_content</item>
        <item name="android:textSize">30sp</item>
        <item name="android:layout_weight">1</item>
        <item name="android:textStyle">bold</item>
        <item name="backgroundTint">@color/black</item>
        <item name="android:layout_marginEnd">10dp</item>
    </style>
</resources>
```

Output:

